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Battery Storage Market Notes, July 2026 issue

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"Grid-scale battery storage" (a large battery connected to the transmission grid that stores electricity and releases it) earns money by selling the "capacity" to correct electricity surpluses and shortages. It also uses the price differences between times of day. These notes use published figures and our own records to show this month's moves in figures and tables. This issue is a back issue, produced retroactively in November 2026. The grid connection study totals are as of the time of production (October 19, 2026).

1. The 3 revenue sources of a battery storage plant (how to read these notes)

A battery storage plant (grid-scale battery) earns revenue from 3 markets. ① The balancing market (EPRX) is where the "capacity" to correct electricity surpluses and shortages is sold to grid operators. ② The wholesale electricity market (JEPX) is where you buy and store electricity at cheap times and sell it at expensive times. ③ The capacity market (OCCTO) is where the government buys future electricity supply capacity in advance. These notes present the figures in the order of these 3 markets. The ①②③ at the start of each section title shows which market it covers.

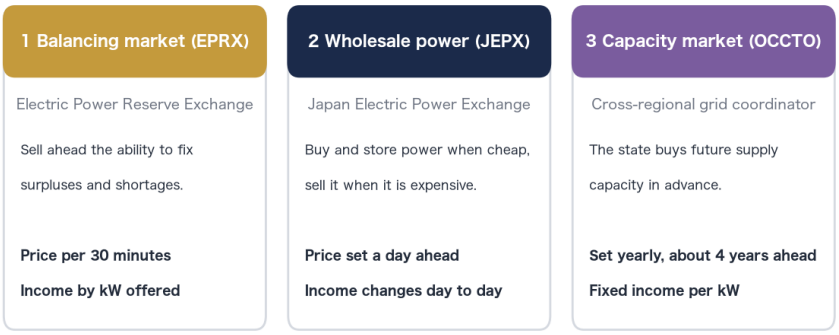


Figure 1 The 3 revenue sources of a battery storage plant

Source: Rules of the Japan Electric Power Exchange (JEPX), the Electric Power Reserve Exchange (EPRX) and the Organization for Cross-regional Coordination of Transmission Operators (OCCTO).

2. Three trends this month

We picked 3 moves this month from the figures of the 3 markets.

- In the balancing market (EPRX), the bid-to-cover ratio of Primary reserve (online) went above 100% at 106.8% in June 2026. This month it was 119.7%, and the average winning price was 4.35 yen/kW per 30 min.
- In the balancing market (EPRX), 57.3% of Tertiary reserve 2 sales in July 2026 came from the top 10% of 30-minute slots by sales.
- In the wholesale electricity market (JEPX), the monthly average spread between the highest 4 hours and the lowest 4 hours in July 2026 was 13.17 yen/kWh in Tokyo and 15.71 yen/kWh in Kyushu.

3. ① Balancing market (EPRX) | 7 products, and prices by area

The balancing market is where grid operators buy, every 30 minutes, the "capacity" to correct electricity surpluses and shortages. Products are divided by how fast they change output (Primary reserve responds to changes in frequency; Secondary and Tertiary reserves respond to dispatch instructions from the grid operator). Primary reserve is the fastest and Tertiary reserve 2 is the slowest. Primary reserve is a product that senses changes in frequency by itself and responds. It has 2 categories: "online", where the grid operator monitors operation over a dedicated line, and "offline", where no dedicated line is laid and a record of operation is submitted afterwards. These are offered and traded separately, so we show them separately. A Composite product is a bid that combines several products. Prices are set in yen per 1 kW of available capacity per 30 minutes. kW is a unit of the size of power that can be delivered, which is different from kWh, the amount of electricity. Each 30-minute trading round is called a "30-minute slot". The figure shows the range of winning prices by area (by the area where the power source is located). The thin line runs from the lowest to the highest winning bid in the month, the thick band is the price range that covers 8 of every 10 30-minute slots with winning bids, and the dot is the average winning price. Nationwide figures are in the table in the "Market figures" section.

By area, the highest average winning price in the balancing market was Primary reserve (offline) in Tohoku at 14.49 yen/kW per 30 min, and the lowest was Tertiary reserve 2 in Shikoku at 0.38 yen/kW per 30 min.



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Figure 2 Range and average of winning prices by area in the balancing market (July 2026 (1st to 31st), by area where the power source is located)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our compilation.

Product	Response time	Duration
Primary reserve (online)	Within 10 seconds	5 minutes or more
Primary reserve (offline)	Within 30 seconds	No set duration
Secondary reserve 1	Within 5 minutes	30 minutes
Secondary reserve 2	Within 5 minutes	30 minutes
Tertiary reserve 1	Within 15 minutes	30 minutes
Tertiary reserve 2	Within 60 minutes	30 minutes
Composite product	Requirements of the combined products	Requirements of the combined products

4. ① Balancing market (EPRX) | Grid areas and times where bids win more easily

The win rate is the share of the bid volume that wins. The higher it is, the more easily bids win. The figure plots the win rate for each product by time of day (every 30 minutes), for each grid area (area where the power source is located). Where a line is broken, there were no bids at that time.

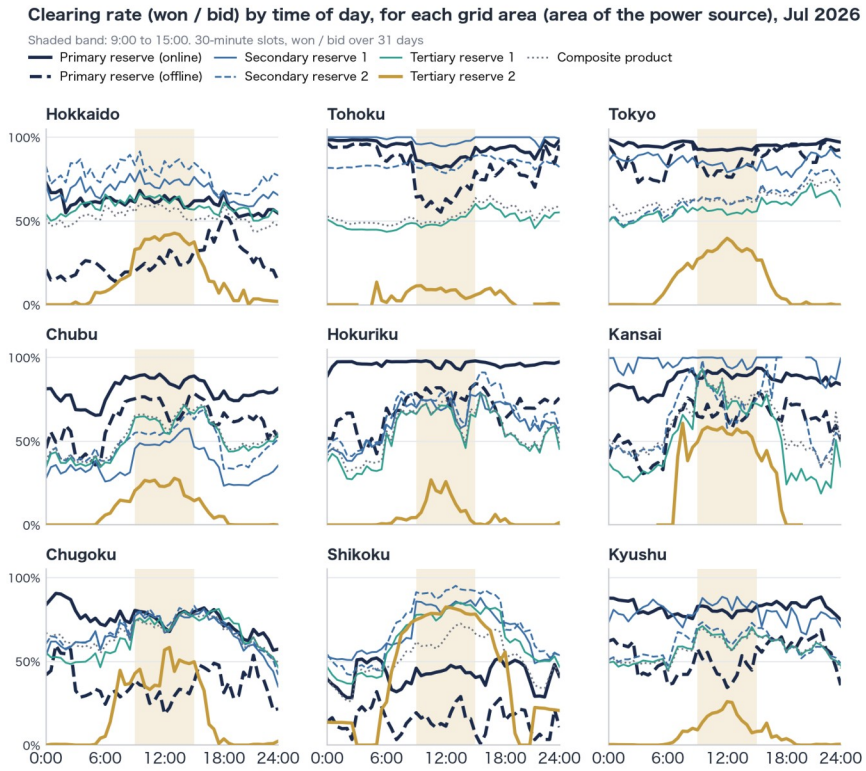


Figure 3 Win rate in the balancing market (grid area (area where the power source is located) × time of day, July 2026 (1st to 31st))

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our compilation.

5. ① Balancing market (EPRX) | Primary reserve (online) became a competition for slots

Primary reserve is the fastest product. It responds to changes in frequency and changes output within 10 seconds (online) or within 30 seconds (offline). Batteries are often used for it. The left side is the online category and the right side is the offline category. The bars show the bid-to-cover ratio (bid volume ÷ volume procured by grid operators). Above 100% means more bids came in than the volume procured. Even below 100%, some bids do not win (see the win rate in the "Market figures" table). The bars rise both when bids increase and when the procured volume falls. The line shows the average winning price. These are monthly figures from April 2025, the data we have.

In the balancing market, the bid-to-cover ratio of Primary reserve (online) went above 100%, at 106.8% in June 2026 and 119.7% this month. Its average winning price was 2.89 yen/kW per 30 min in April 2025 and 4.35 yen/kW per 30 min this month. Primary reserve (offline) had a bid-to-cover ratio of 65.1% and an average winning price of 10.25 yen/kW per 30 min this month. From April 2025 to this month, the procured volume fell from 3,066.1 MW to 1,408.5 MW for online and from 1,122.5 MW to 495.7 MW for offline.

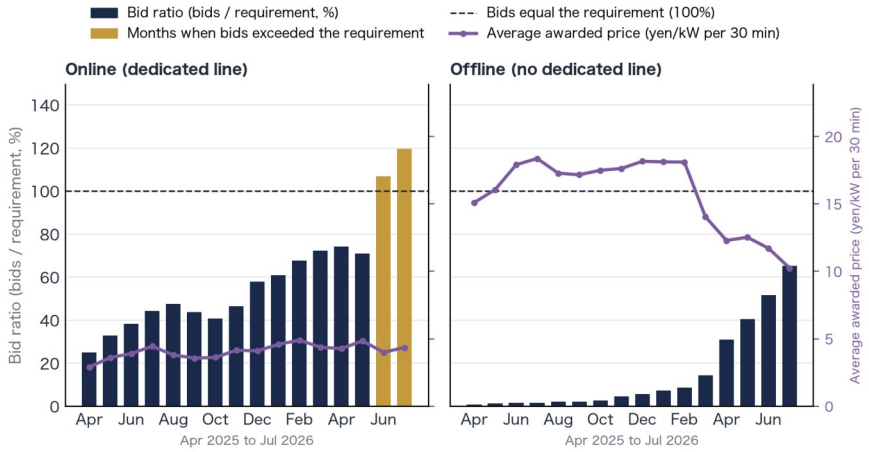


Figure 4 Bid-to-cover ratio (bars) and average winning price (line) of Primary reserve (monthly; online on the left, offline on the right; gold bars are months when bids exceeded the procured volume)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our compilation.

6. ① Balancing market (EPRX) | How prices by product have moved

These are the average winning prices by product from April 2025 (monthly averages weighted by winning volume). From March 14, 2026, Primary to Tertiary reserve 1 and the Composite product moved from trading in 3-hour blocks to trading every 30 minutes (Tertiary reserve 2 was already traded every 30 minutes).

In the balancing market, the average winning price of Tertiary reserve 2 was 1.12 yen/kW per 30 min in April 2025 and 1.88 yen/kW per 30 min this month, with a high of 4.18 yen/kW per 30 min in May 2026. Primary reserve (offline) was 15.10 yen/kW per 30 min in April 2025 and 10.25 yen/kW per 30 min this month, with a high of 18.37 yen/kW per 30 min in July 2025.

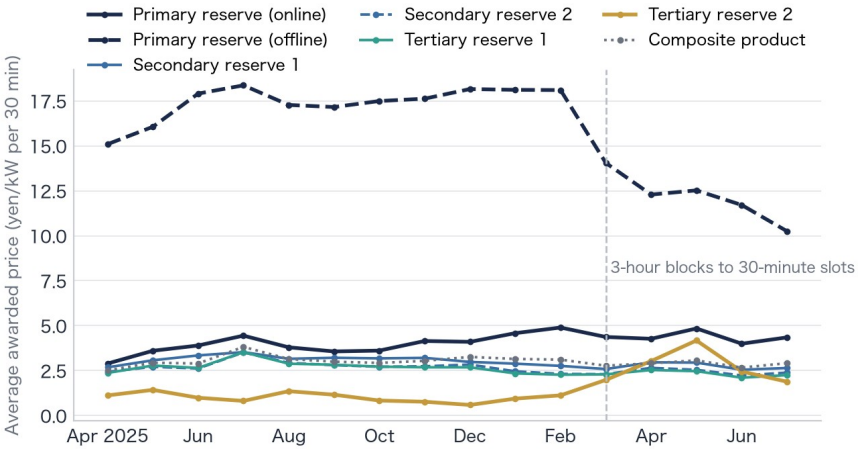


Figure 5 Average winning price by product in the balancing market (yen/kW per 30 min, monthly)
Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our compilation.

7. ① Balancing market (EPRX) | Which times of day are expensive

These are the hourly average winning prices by product for July 2026 (1st to 31st).

In the balancing market, the average winning price of Primary reserve (online) was highest around 18:00 at 6.07 yen/kW per 30 min and lowest around 3:00 at 3.44 yen/kW per 30 min. Tertiary reserve 2 was highest around 20:00 at 17.85 yen/kW per 30 min and lowest around 2:00 at 0.37 yen/kW per 30 min.

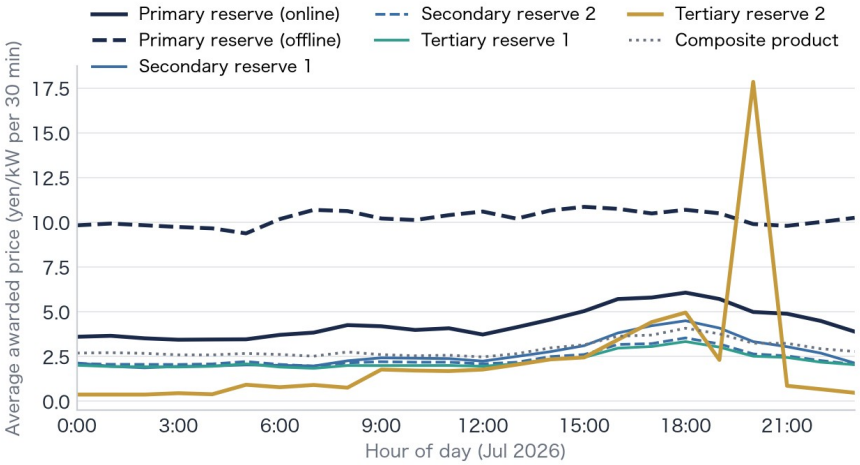


Figure 6 Average winning price by time of day in the balancing market (July 2026 (1st to 31st), yen/kW per 30 min)
Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our compilation.

8. ① Balancing market (EPRX) | Which products batteries are used for

The figure shows the share of winning volume held by batteries. The table compares the average winning price of the volume won by batteries with the average winning price of the product as a whole.

In the balancing market, batteries held 22.8% of Primary reserve (online), and the average winning price of the volume won by batteries was 8.50 yen/kW per 30 min (overall 4.35 yen/kW per 30 min). Batteries held 100.0% of Primary reserve (offline), and both the battery volume and the overall price were 10.25 yen/kW per 30 min.

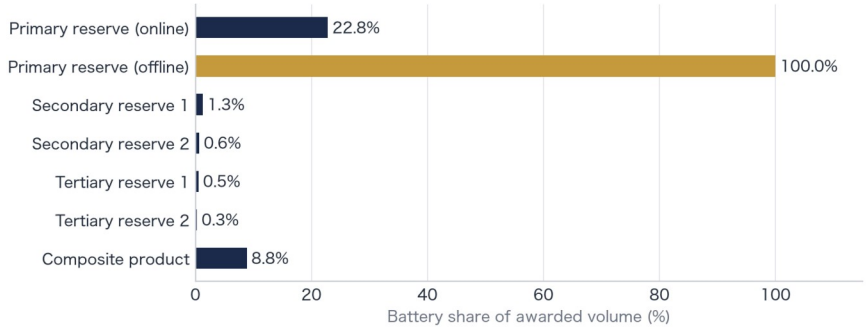


Figure 7 Share of batteries in the balancing market (%)

Source: Trading results (preliminary) by power source type, published by the Electric Power Reserve Exchange (EPRX). Our compilation.

Product	Volume won by batteries (yen/kW per 30 min)	Product overall (yen/kW per 30 min)
Primary reserve (online)	8.50	4.35
Primary reserve (offline)	10.25	10.25
Secondary reserve 1	5.52	2.65
Secondary reserve 2	4.47	2.39
Tertiary reserve 1	4.43	2.25
Tertiary reserve 2	51.97	1.88
Composite product	8.48	2.91

Both are averages weighted by winning volume (July 2026, preliminary, nationwide). The battery share of Tertiary reserve 2 is only 0.3% of winning volume. The nationwide average winning price of the 30-minute slots where batteries won, weighted by battery winning volume, is 7.02 yen. The battery average is high not because batteries are concentrated in high-priced slots, but because the battery bids themselves are priced high (each winning bid is settled at its own bid price).

9. ① Balancing market (EPRX) | Sales are concentrated in a small number of slots

We ranked the month's 30-minute slots with winning bids by sales (average winning price × winning volume, nationwide), from highest to lowest, and plotted for each product what % of the month's sales the top slots account for. The more a curve bulges above the dotted line, the more sales are concentrated in fewer slots. Tertiary reserve 2 is a product that covers forecast errors in weather-dependent renewable generation. Its price is low in most slots and high in only a few.

In the balancing market, the top 10% of slots accounted for 57.3% of July 2026 sales for Tertiary reserve 2 and 17.7% for Primary reserve (online). The top 1% of slots accounted for 11.4% for Tertiary reserve 2 and 2.2% for Primary reserve (online).

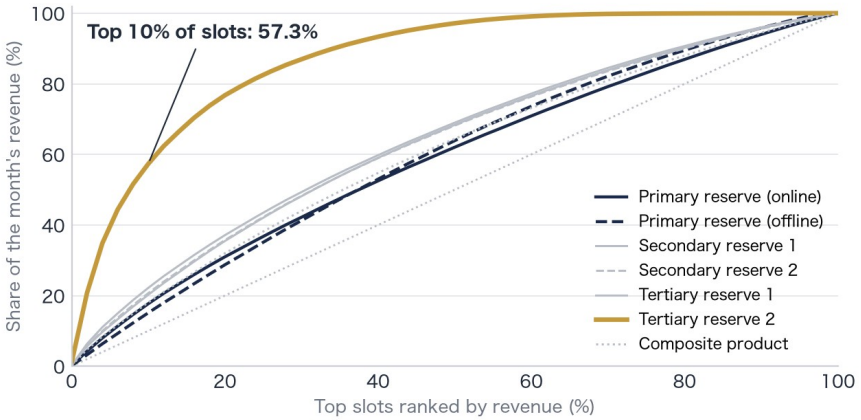


Figure 8 Share of balancing market sales concentrated in the top slots (July 2026 (1st to 31st), nationwide)

Source: Trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our compilation.

In July 2026, out of 1,188 30-minute slots with winning bids, the top 1% by sales (average winning price × winning volume) accounted for 11.4% of the month's sales. There were 39 slots with a nationwide average winning price of 10 yen/kW per 30 min or more, and 300 slots with no winning bids.

Slots with the highest maximum winning prices (by area where the power source is located). The maximum winning price is the price of the single highest winning bid. The average winning price for the same slot and area is shown on the right:

Date	Time	Area	Maximum winning price (yen/kW per 30 min)	Average winning price for the same slot and area (yen/kW per 30 min)
July 11	9:00	Tokyo	97.00	16.43
July 11	9:30	Tokyo	97.00	16.28
July 11	10:00	Tokyo	97.00	16.51
July 11	10:30	Tokyo	97.00	10.00
July 15	9:00	Tokyo	97.00	16.09

Source: Tertiary reserve 2 trading results (preliminary) published by the Electric Power Reserve Exchange (EPRX). Our compilation.

10. ② Wholesale electricity (JEPX) | Average prices

Wholesale electricity is electricity bought and sold between power companies and retailers. Its price is set on the "spot market" of the Japan Electric Power Exchange (JEPX) (a market where electricity for the next day is traded the day before). "System (nationwide)" is the price when all of Japan is treated as one market (the system price). When the transmission lines between regions become congested, prices split by area from Hokkaido to Kyushu (area prices).

The monthly average of the System (nationwide) price on the wholesale market (JEPX) in July 2026 was 17.91 yen/kWh, compared with 12.75 yen/kWh in the same month last year.

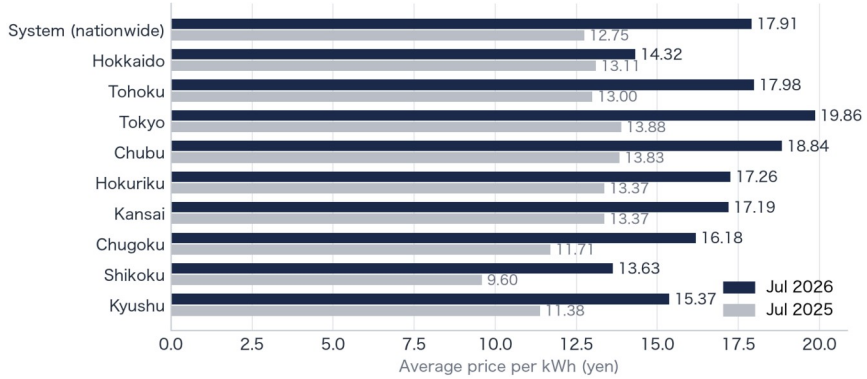


Figure 9 Average wholesale electricity price (yen/kWh). Navy is July 2026, grey is the same month last year

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Simple average of 30-minute prices. Our compilation.

11. ② Wholesale electricity (JEPX) | How the price moves through the day

The wholesale electricity price changes every 30 minutes. The figure shows the daily shape, averaging the price at the same time of day across the month. If a battery charges at cheap times and discharges at expensive times, the difference becomes its revenue.

The daily low and peak of the wholesale price were 15.28 yen/kWh around 6:00 and 28.07 yen/kWh around 18:30 in Tokyo, and 9.85 yen/kWh around 12:30 and 27.02 yen/kWh around 18:30 in Kyushu.

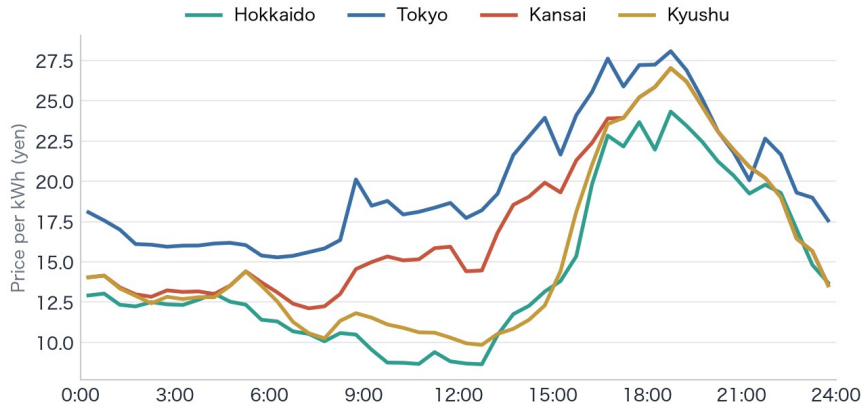


Figure 10 Daily peaks and lows of the wholesale electricity price (July 2026 average, yen/kWh)

Source: JEPX spot market. Our compilation.

12. ② Wholesale electricity (JEPX) | How much is the spread per kWh

For each area, we took the average price of the day's highest 4 hours minus the average price of the day's lowest 4 hours, and show the monthly average. This equals the difference per kWh if you buy 1 kWh in the cheapest 4 hours and sell it in the most expensive 4 hours. It is a simple difference: it does not deduct electricity lost in charging and discharging or fees, and it ignores the order of the cheap and expensive hours.

The monthly average spread between the highest 4 hours and the lowest 4 hours of the wholesale price was largest in Shikoku at 16.78 yen/kWh and smallest in Tokyo at 13.17 yen/kWh.

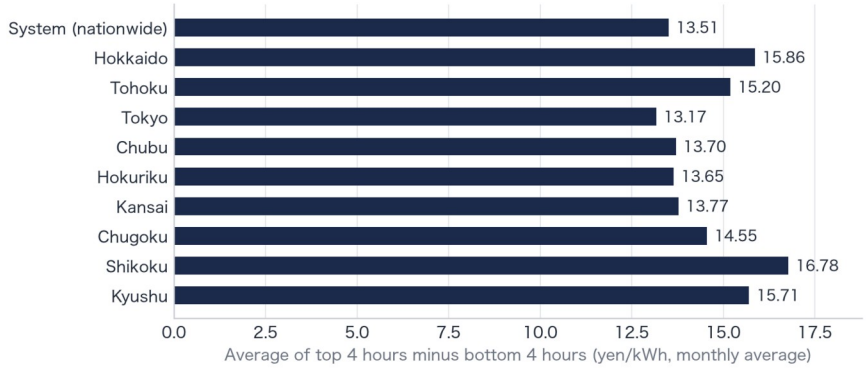


Figure 11 Wholesale electricity spread: average of the highest 4 hours − average of the lowest 4 hours (July 2026, yen/kWh, monthly average)

Source: JEPX spot market. Our compilation.

Area	2-hour spread (yen/kWh)	4-hour spread (yen/kWh)	Days with a 4-hour spread of 10 yen or more
System (nationwide)	14.79	13.51	21
Hokkaido	17.92	15.86	25
Tohoku	17.24	15.20	26
Tokyo	15.15	13.17	22
Chubu	14.98	13.70	23
Hokuriku	14.98	13.65	21
Kansai	15.20	13.77	21
Chugoku	16.09	14.55	21
Shikoku	18.97	16.78	23
Kyushu	17.04	15.71	22

The spread is the monthly average of the daily differences. July 2026 covers 31 days.

13. ② Wholesale electricity (JEPX) | There is a spread every day

We show the same 4-hour spread day by day. The larger the spread on a day, the more room a battery has to earn from it.

Out of 31 days, the 4-hour wholesale spread was 10 yen or more on 21 days for System (nationwide), and on 26 days in Tohoku, the area with the most.

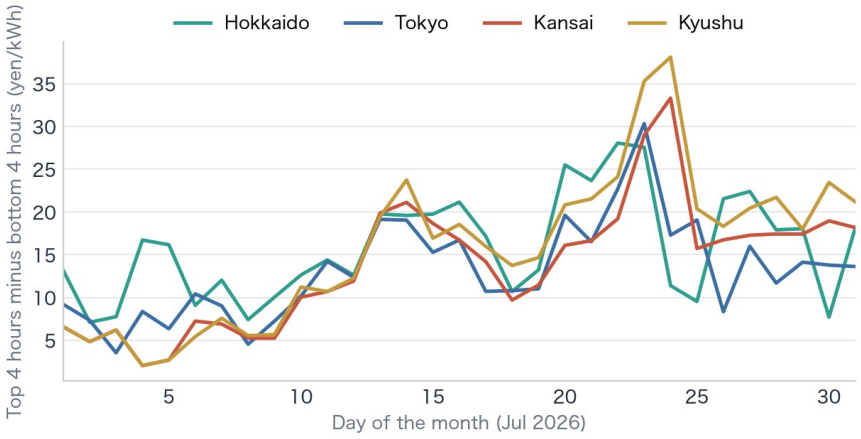


Figure 12 Daily wholesale electricity spread (average of the day's highest 4 hours − average of the lowest 4 hours, yen/kWh)

Source: JEPX spot market. Our compilation.

14. ③ Capacity market (OCCTO) | A market that buys future supply capacity

The capacity market buys electricity supply capacity 4 years ahead through an annual auction. Prices are set in yen per kW per year. The left figure shows the clearing prices of past main auctions. The right figure shows the battery and pumped hydro capacity awarded in the Long-Term Decarbonization Power Source Auction (a separate auction that promises capacity revenue, in principle for 20 years, for new decarbonized power sources and similar). These are published once a year, so they do not change every month. The latest is delivery fiscal year FY2029 (published January 2026).

The capacity market clearing price for Tokyo was 14,137 yen/kW for delivery FY2024 and 3,495 yen/kW for FY2025. The latest publication, which comes once a year and does not change monthly, is 15,111 yen/kW for FY2029. In the latest publication of the Long-Term Decarbonization Power Source Auction (FY2025 round), also once a year, the awarded battery and pumped hydro capacity was 819 MW for lithium-ion batteries and pumped hydro (excluding new build), and 886 MW for non-lithium-ion batteries, pumped hydro (new build) and LDES.

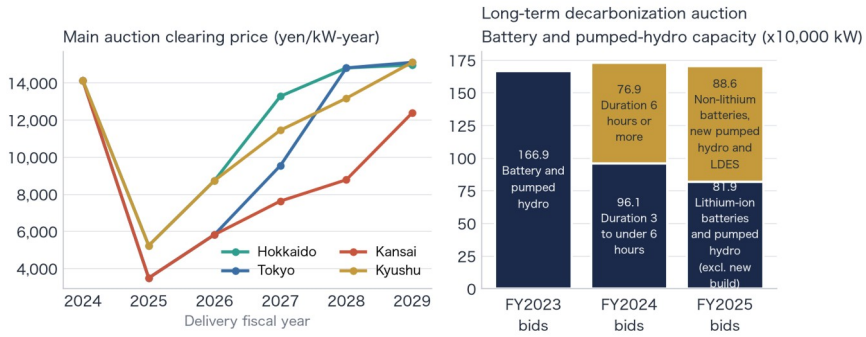


Figure 13 Capacity market clearing prices (left, yen/kW) and battery and pumped hydro capacity awarded in the Long-Term Decarbonization Power Source Auction (right, in units of 10,000 kW)

Source: Clearing results and capacity contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in the last section (Assumptions and terms)). As published.

15. Our records | Where battery storage projects are increasing

We counted, by grid area of the grid operator, the projects newly listed this month on DealRoom (our data room for buyers), a site where we list battery storage projects for buyers. These are figures for projects we handled, not counts for the whole market. Transfer of rights is a sale that hands over the land and grid rights. Completed-build handover is a sale that hands over the finished facility.

New listings on DealRoom (our data room for buyers) in July 2026 were 27 projects, 137.3 MW. By grid area, these included Tokyo with 12 projects, 23.9 MW, Chugoku with 4 projects, 91.4 MW, and Kyushu with 5 projects, 10.0 MW.

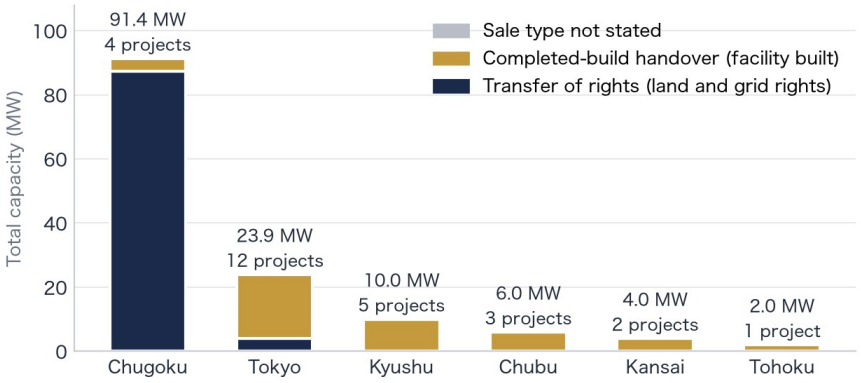


Figure 14 Capacity of projects we listed on DealRoom (our data room for buyers) in July 2026 (MW)

Source: Records of DealRoom (our data room for buyers) (project locations are shown only to the grid area level; prices and parties are not listed).

16. Our records | How many months and how much it costs to connect to the grid

A grid connection study is a procedure in which you ask the grid operator to check "can we connect at this site, and how much will the work cost". The grid connection cost contribution is the part of the connection work cost that the applicant pays. From the response documents we received, we compiled only the high-voltage ones (connecting to 6,600 V distribution lines) by grid area. Extra-high voltage is not included. Grid areas with fewer than 5 cases are not shown.

The medians of grid connection studies (high voltage only, as of October 19, 2026) range from 8.0 months (Tokyo) to 17.0 months (Tohoku) for the construction period, and from ¥3.02 million (Tokyo) to ¥32.35 million (Tohoku) for the grid connection cost contribution.

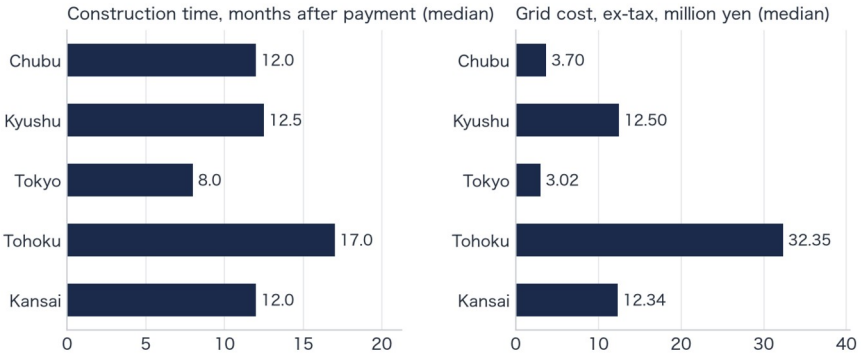


Figure 15 Median of grid connection study responses (high voltage) by grid area (left: months of construction work; right: grid connection cost contribution, excluding tax)

Source: Our compilation of the high-voltage grid connection study response documents we received from general transmission and distribution operators (as of October 19, 2026).

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17. Market figures (tables ready to use in your materials)

We put the figures behind the charts into tables so you can use them in your materials: ① nationwide figures by product for the balancing market (bid-to-cover ratio, win rate, price, battery share), ② figures by area and trading volume for the wholesale market (JEPX), and ③ capacity market clearing prices.

Balancing market (July 2026, preliminary, nationwide)

Product	Procured volume (MW)	Bid volume (MW)	Winning volume (MW)	Bid-to-cover ratio (bids ÷ procured)	Win rate (winning ÷ bids)	Average winning price (yen/kW per 30 min)	Battery share
Primary reserve (online)	1,408.5	1,685.7	1,334.8	119.7%	79.2%	4.35	22.8%
Primary reserve (offline)	495.7	322.7	207.7	65.1%	64.4%	10.25	100.0%
Secondary reserve 1	924.0	1,972.7	1,190.9	213.5%	60.4%	2.65	1.3%
Secondary reserve 2	831.0	3,496.6	2,139.8	420.8%	61.2%	2.39	0.6%
Tertiary reserve 1	2,455.6	5,280.7	2,922.3	215.0%	55.3%	2.25	0.5%
Tertiary reserve 2	333.0	1,561.8	329.6	469.0%	21.1%	1.88	0.3%
Composite product	3,290.1	5,975.8	3,459.0	181.6%	57.9%	2.91	8.8%

Volumes are averages per 30-minute slot. Products with a bid-to-cover ratio below 100% did not receive enough bids for the volume procured. The win rate is the share of the bid volume that won.

Source: Trading results (preliminary) and trading results by power source type, published by the Electric Power Reserve Exchange (EPRX). Our compilation.

Wholesale electricity, JEPX spot market (July 2026, by area, yen/kWh)

Area	Monthly average	Same month last year	Daytime (8:00–16:00)	Evening (16:00–20:00)	4-hour spread	Maximum	Slots at 30 yen or more	Slots at 0.01 yen
System (nationwide)	17.91	12.75	15.97	26.02	13.51	46.88	137	0
Hokkaido	14.32	13.11	10.57	22.58	15.86	45.01	67	6
Tohoku	17.98	13.00	15.06	25.96	15.20	50.01	102	3
Tokyo	19.86	13.88	19.75	26.70	13.17	50.01	150	0
Chubu	18.84	13.83	17.61	26.38	13.70	46.40	160	0
Hokuriku	17.26	13.37	16.67	24.78	13.65	46.40	151	0
Kansai	17.19	13.37	16.48	24.89	13.77	58.38	147	0
Chugoku	16.18	11.71	13.55	24.71	14.55	58.38	123	0
Shikoku	13.63	9.60	9.89	20.64	16.78	58.38	100	23
Kyushu	15.37	11.38	11.60	24.68	15.71	58.38	123	0

Trading volume (July 2026, total of 31 days): total contracted volume 37,697 million kWh (same month last year 27,205 million kWh), sell bids 55,527 million kWh, buy bids 46,604 million kWh.

Source: Spot market trading results published by the Japan Electric Power Exchange (JEPX). Prices are simple averages of 30-minute slots. Our compilation.

Capacity market main auction area prices (yen/kW per year)

Delivery fiscal year	Hokkaido	Tohoku	Tokyo	Chubu	Hokuriku	Kansai	Chugoku	Shikoku	Kyushu
FY2024	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137	14,137
FY2025	5,242	3,495	3,495	3,495	3,495	3,495	3,495	3,495	5,242
FY2026	8,749	5,833	5,834	5,832	5,832	5,832	5,832	5,832	8,748
FY2027	13,287	9,044	9,555	7,823	7,638	7,638	7,638	7,638	11,457
FY2028	14,812	14,812	14,812	10,280	8,785	8,785	8,785	8,785	13,177
FY2029	14,972	15,111	15,111	12,388	12,388	12,388	12,388	12,388	15,112

Battery and pumped hydro capacity awarded in the Long-Term Decarbonization Power Source Auction (10,000 kW)

Auction round	Category	Awarded capacity (10,000 kW)
FY2023 round	Batteries and pumped hydro	166.9
FY2024 round	Continuous operation of 3 hours to under 6 hours	96.1
FY2024 round	Continuous operation of 6 hours or more	76.9

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Auction round	Category	Awarded capacity (10,000 kW)
FY2025 round	Lithium-ion batteries and pumped hydro (excluding new build)	81.9
FY2025 round	Non-lithium-ion batteries, pumped hydro (new build) and LDES	88.6

Source: Clearing results and capacity contract documents published by the Organization for Cross-regional Coordination of Transmission Operators (OCCTO) (sources listed in the last section (Assumptions and terms)). As published.

18. Assumptions and terms

- The figures are our compilation of the published values from the sources. We do not add up several months or convert to annual values. We do not show revenue amounts for any single battery storage plant.
- Period covered: for the balancing market (EPRX), preliminary values for July 2026 (delivery dates up to the 31st); for wholesale electricity (JEPX), July 2026 (31 days). The month for the balancing market differs until the published values for July 2026 are included in our compilation.
- Balancing market: nationwide (total column) of EPRX preliminary values. Procured volume is by grid operator (TSO); bid volume, winning volume and average winning price are by area where the power source is located. Volumes are averages per 30-minute slot. The average winning price is weighted by winning volume. Bid-to-cover ratio = bid volume ÷ procured volume; win rate = winning volume ÷ bid volume. The 3-hour blocks up to March 13, 2026 were counted as 6 30-minute slots per block.
- Range of winning prices: the thin line runs from the lowest bid (minimum winning price) to the highest bid (maximum winning price) among the month's 30-minute slots with winning bids (published values by area where the power source is located). The published values for Primary reserve (online), Secondary reserve 1 and 2, and Tertiary reserve 1 also include Composite product bids (one price for several products combined). So we removed from the thin line any bids above the price cap of 15 yen for Primary reserve (online) and Secondary reserve 1 (for deliveries up to August 31, 2026), and above 10 yen for Secondary reserve 2 and Tertiary reserve 1, where a Composite product bid in the same slot and area can explain them. For Primary reserve (offline), we removed from the thin line bids above the price cap (15 yen this month) (the balancing market system also clears bids above the cap and lowers them to the cap at settlement). The thick band is the range from the bottom 10% to 90% when the average winning prices of the slots with winning bids are sorted from lowest to highest. The dot is the average weighted by winning volume.
- Win rate by grid area × time of day: total winning volume ÷ total bid volume for slots at the same time of day in the month (by area where the power source is located).
- The response time and duration of the balancing market products are the FY2026 specifications (Electric Power Reserve Exchange, "Product requirements", 6th edition). Primary reserve senses changes in frequency by itself and responds. Online is the category where the grid operator monitors operation over a dedicated line; offline is the category where no dedicated line is laid and a record of operation is submitted afterwards. The offline response time is within 30 seconds, and there is no set duration.
- High-priced 30-minute slots for Tertiary reserve 2: slots with a nationwide average winning price of 10 yen/kW per 30 min or more. The top 1% means the top 1% of slots ranked by sales (average winning price × winning volume) (1% of the number of slots, rounded). The maximum winning price is the published value by area where the power source is located, and is the price of the single highest winning bid. In the balancing market, each winning bid is settled at its own bid price (pay-as-bid (multi-price)), so the average winning price for the same slot and area is lower than this.
- Average prices are simple averages of 30-minute slots (not weighted by trading volume). Daytime is the slots from 8:00 to 16:00; evening is the slots from 16:00 to 20:00.
- Spread: the day's 30-minute slots are sorted by price, and the average of the 8 lowest slots is subtracted from the average of the 8 highest slots (4 hours) (4 slots for 2 hours). It does not consider the order of cheap and expensive times, electricity lost in charging and discharging, or fees, taxes and grid usage charges. It does not represent actual revenue.

Sources (OCCTO, capacity market clearing results): delivery FY2024
https://www.occto.or.jp/iinkai/youryou/kentoukai/2020/files/youryou_kentoukai_27_05.pdf / delivery FY2025
https://www.occto.or.jp/iinkai/youryou/kentoukai/2021/files/youryou_kentoukai_35_03_1.pdf / delivery FY2026
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Sources (OCCTO, Long-Term Decarbonization Power Source Auction): FY2023 round https://www.occto.or.jp/assets/260605_auction_keiyakujyoukyou_kouhyou.pdf / FY2024 round https://www.occto.or.jp/market-board/market/jitsujukyukanren/files/longauction_keiyakukekka_2024.pdf / FY2025 round https://www.occto.or.jp/assets/various/capacity-market/jitsujukyukanren/yoryokakuhokeiyaku/longauction_keiyakukekka_2025.pdf

- Capacity market clearing results for each delivery fiscal year are published in different years (FY2024 in 2020, FY2029 in January 2026). The number of battery projects in the Long-Term Decarbonization Power Source Auction is not shown because it is not included in the published summary tables.

Sources: Japan Electric Power Exchange (JEPX) spot market trading results / Electric Power Reserve Exchange (EPRX) trading results (preliminary) and trading results by power source type / Organization for Cross-regional Coordination of Transmission Operators (OCCTO) capacity market main auction clearing results and Long-Term Decarbonization Power Source Auction capacity contract documents / our compilation of grid connection study response documents we received / records of DealRoom (our data room for buyers).

Terms: System price (System, nationwide) = the wholesale electricity price when all of Japan is treated as one market. Area price = the price that splits by area depending on congestion on transmission lines. The grid = the power companies' transmission network. Grid connection = connecting to the transmission network. MW = size of power (1 MW = 1,000 kW). MWh = amount of electricity that can be stored. yen/kWh = price per 1 kWh of electricity.

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